Anderson Junior College
H1 Chemistry 8873 **2018 JC2 Prelim Paper 2 Suggested Solutions**

1 **(a)** Standard enthalpy change of combustion is the heat evolved when one mole of substance, in its standard state, is burnt in excess oxygen under standard conditions of 298 K and 1 bar. [2]

[1]: "heat evolved", "one mole", "excess oxygen"
[1]: 298 K and 1 bar

(b) ($CH_3CH_2OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$)

Heat absorbed by water (65% efficient) = $mc\Delta T$
= (100)(4.18)(65 – 20)
= 18810 J [1]

Heat evolved by combustion of ethanol (100%) = $\dfrac{18810}{0.65}$
= 28938 J
= 28.94 kJ [1]

Enthalpy change of combustion of ethanol = $-\dfrac{28.94}{\frac{1}{46}}$
= –1330 kJ mol^{-1} [1]

(c) **(i)** (standard) enthalpy change of formation of ethanol [1]

(ii) [3]

$$\begin{tikzcd}
2C(s) + 3H_2(g) + \frac{7}{2}O_2(g) \arrow[rr, "\Delta H_1"] \arrow[dr, "\Delta H_3"'] & & CH_3CH_2OH(l) + 3O_2(g) \arrow[dl, "\Delta H_2"] \\
& 2CO_2(g) + 3H_2O(l) &
\end{tikzcd}$$

$\Delta H_1 = \Delta H_3 - \Delta H_2$
= [2(–393.5) + 3(–285.8)] – (–1330)
= –1644.4 + 1330
= –314 kJ mol^{-1}

[1]: application of Hess' Law
[1]: correct answer with sign (ecf)
[1]: correct units

(d) **(i)** 6 453 600 x 159 x 2.33 = 2.39 x 10^9 kg (3 s.f.) [1]

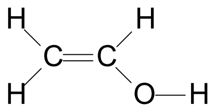
(ii) 1/100 x 2.33 = 0.0233 kg [1]

(iii) 0.1 x 0.0233 + 0.9 x 2.33 = 2.10 kg (3 s.f.) or 2.099 kg [1]

(iv) 6 453 600 x 159 x 2.10 / 1000 = 2.15 x 10^9 kg (3 s.f.) [1]

Reduction in CO_2 emission = 2.39 x 10^9 – 2.15 x 10^9 = 2.40 x 10^8 kg

Alternative
[(2.33 – 2.10) x 6 453 600 x 159] kg

(e) **(i)**  [1]

(ii) H_2 with Ni at high temperature and pressure [1]
or
H_2 with Pt / Pd, r.t.p

(iii) Hydrogen bonding [1]

(iv) [1]: Partial charges on O–H bond of PVA + O–H bond of water [2]
[1]: Show attraction between lone pair on O atom with H atom of O–H group

(v) The borate ion forms cross linkages via hydrogen bonds between PVA polymer molecules. [1]

(vi) dilute H_2SO_4, heat (accept NaOH(aq)) [1]

2 (a) A thermoplastic is a substance / polymer that softens on heated and hardens when cooled. [1]

(b) (i) LDPE has branched polymer chains; [1]
less close / less regular packing so weaker instantaneous dipole – induced dipole attraction [1]

HDPE has no / little branching in polymer chains; [1]
closer and regular packing so stronger instantaneous dipole – induced dipole attraction [1]

(ii) LDPE: plastic carrier bags or other similar low strength and flexible sheet materials [1]

HDPE: milk bottles and similar containers, washing up bowls, plastic pipes [1]

(c) (i) X: I_2 or iodine
Y: Cl_2 or chlorine (accept F_2 or fluorine)
Z: Br_2 or bromine

[2]: all correct
[1]: 1 or 2 correct

(ii) Chlorine, Cl_2 (accept Y_2) [1]

Chloride or Y^- not accepted

(iii) Although chlorine has the smallest nuclear charge, it has the least shielding effect / least number of electron shells. [1]

Valence shell of chlorine atom is closest to the nucleus hence the incoming electron that resides in it will be more strongly attracted / easily gained. [1]

(d) (i) Thermal stability: $HCl > HBr > HI$ [1]

(ii) Down the group, H–X bond energy decreases from 431 to 299 kJ mol^{-1}. [1]

Reason:
- size of halogen atom increases
- bonding electrons are further from both nuclei
- and less strongly attracted to the two nuclei
- (H–X bond strength decreases)

[1]

(e) (i) Similarity: both are electrical conductors [1]

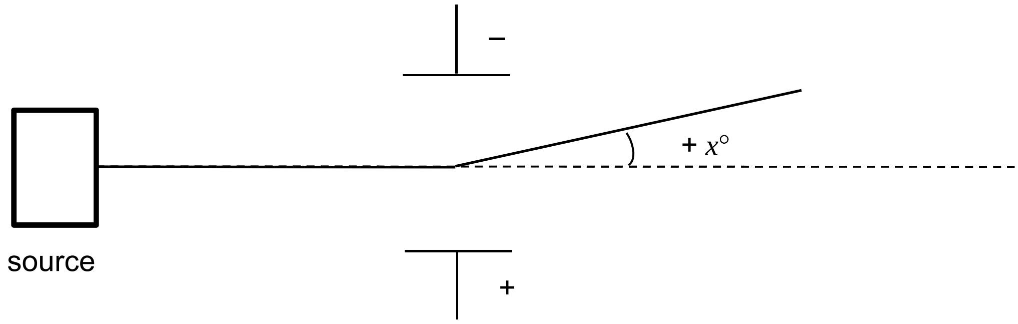
Difference: CNT is rigid but graphene is flexible. [1]

(ii) A dimension between 1 – 100 nm [1]

(iii) Via digestive tract / oral uptake [1]

Via the skin / dermal uptake [1]

3 (a) (i) [1]



The plate on top is negatively charged as the positively charged proton is attracted to it. [1]

(ii) **I**: $^{16}O^{2-}$ ions

q/m of proton = 1/1 [1]

q/m of $^{16}O^{2-}$ ions = 2/16 = 1/8 [1]

Hence, angle of deflection is 1/8 that of proton, and in the opposite direction / (towards positive plate),

$-x/8°$.

II: ^{4}He nuclei

q/m = 2/4 = ½ [1]

Hence, angle of deflection is 1/2 that of proton, $+x/2°$ (towards negative plate) [1]

III: ^{2}H atom

0°. The atom has no net charge and hence remains undeflected. [1]

(b) (i) **1s 2s 2p 3s 3p 4s 3d** [1]

(ii) **$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^1$** [1]

(iii) **4p** [1]

4 **(a)** n(acid) in 250 cm^3 = 2.29 / 104 = <u>0.0220 mol</u> [1]

n(acid) in 25 cm^3 = 0.0220 / 10 = <u>0.00220 mol</u> [1]

(b) Vol of NaOH needed for complete neutralisation = 22.00 cm^3 [1]

n(NaOH) = 22/1000 × 0.100 = <u>0.00220 mol</u>

(c) Since <u>n(acid) = n(NaOH)</u> [1]

It is a <u>monobasic</u> acid.

(d) $[H^+] = 10^{-2.2} = \underline{6.31 \times 10^{-3} \text{ mol dm}^{-3}}$ [2]

$[\text{acid}] = (2.29/104) / (250/1000) = \underline{0.0881 \text{ mol dm}^{-3}}$

Since <u>$[H^+]$ < [acid]</u>, It only partially ionises in water, thus it is a weak acid.

[1]: $[H^+]$ and [acid]
[1]: link $[H^+]$ < [acid] to strength

(e) $K_a = \dfrac{(6.31 \times 10^{-3})^2}{0.0881 - 6.31 \times 10^{-3}}$ (accept $\approx \dfrac{(6.31 \times 10^{-3})^2}{0.0881}$) [2]

$= \underline{4.5 \times 10^{-4} \text{ mol dm}^{-3}}$

Read from graph, when 11 cm^3 of NaOH(aq) was added, pH = pK_a (MBC occurs)
pK_a = 4.0 (accept 3.8 – 4.0)
$K_a = 1.0 – 1.6 \times 10^{-4}$

[1]: use K_a expression in working / recognise pH = pK_a at MBC
[1]: correct answer

(f) Volume of excess NaOH added = 45 – 22 = 23 cm^3 [1]
Amount of excess NaOH = 23/1000 x 0.100 = 2.30×10^{-3} mol

Total volume of solution = 25 + 45 = 70 cm^3
$[OH^-]$ = 0.0023/(70/1000)
= <u>0.0329 mol dm^{-3}</u>

(g) $[H^+][OH^-] = 1 \times 10^{-14}$
$[H^+] = (1 \times 10^{-14}) / 0.0329 = 3.04 \times 10^{-13}$ mol dm^{-3}

pH = – log (3.04×10^{-13}) = <u>12.5</u> [1]

(alternatively, find pOH and then 14 – pOH to get the pH)

5 **(a)** **(i)** Na_2O, dissolves readily in water to give an alkaline solution of pH 13 [1]

P_4O_{10} dissolves readily in water to give an acidic solution of pH 2. [1]

$Na_2O + H_2O \longrightarrow 2NaOH$ [1]

$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$ [1]

(ii) [2]

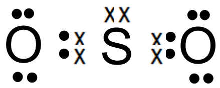
A	B	C
Al_2O_3	P_4O_{10}	Na_2O

1 mark for 1 correct identity
2 mark for 3 correct identity

(iii) Both Na_2O and Al_2O_3 have giant ionic structures with strong electrostatic forces of attraction between the oppositely charged ions. [1]

Since Al^{3+} has a higher charge and a smaller ionic radius than Na^+, [1]

the electrostatic attraction between Al^{3+} and O^{2-} is stronger. Hence more energy is required to break the stronger ionic bond resulting in a higher melting point for Al_2O_3. [1]

(b)  [1]

(c) **(i)** It acts as a catalyst that speeds up the forward and backward reaction such that the equilibrium can be achieved faster. [1]

Is a drying agent and remove water and shift position of equilibrium to the right. Yield of ester increases. [1]

(ii) $K_c = \dfrac{[CH_3CO_2CH_2CH_3][H_2O]}{[CH_3CO_2H][CH_3CH_2OH]}$ [1]

(iii)

	CH_3COOH (l) +	CH_3CH_2OH (l) $\rightleftharpoons$	$CH_3COOCH_2CH_3$ (l) +	H_2O (l)
I / mol	x	x	0	0
C / mol	–1	–1	+1	+1
E / mol	x – 1	x–1	1	1

Let the volume of the reaction mixture be V dm^3

$$4 = \frac{\frac{1}{V} \times \frac{1}{V}}{\frac{x-1}{V} \times \frac{x-1}{V}}$$ [1]

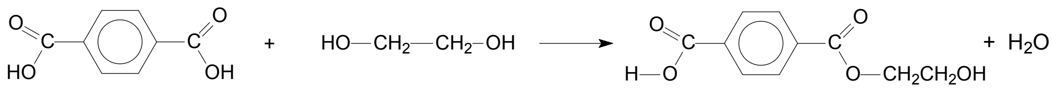
4 = 1 / (x–1)(x–1)

(x–1)(x–1) = ¼
x –1 = ½
x = 1.5 mol
n(ethanoic acid) = 1.5 mol [1]

(iv) Amount of ethyl ethanoate would <u>decrease</u>. [1]

When sodium hydroxide is added, an <u>acid–base reaction</u> would take place between sodium hydroxide and ethanoic acid. This decreases the concentration of ethanoic acid which will <u>shift the POE to the left</u> to increase the concentration of ethanoic acid. [1]

(d) **(i)**



[1]

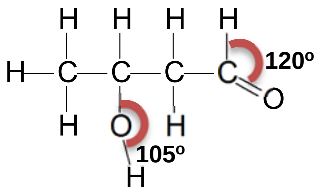
(ii) PET contains <u>ester bonds</u> which will be <u>hydrolysed</u> in the presence of alkaline solution. [1]

Poly(propene) contains <u>non polar C–C and C–H bonds</u> which are <u>inert</u> to alkaline solution. [1]

6 (a) (i) Addition [1]

Two molecules of ethanal react together to form one molecule of 2–hydroxybutanal. [1]

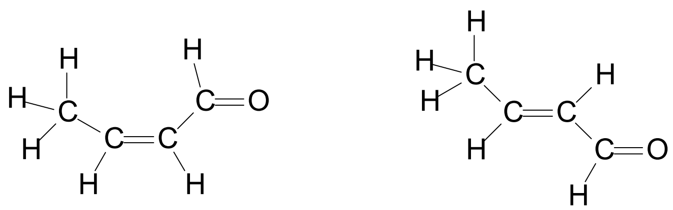
(ii)



Correct displayed formula [1]
Correct 120º bond angle [1]
Correct 105º bond angle [1]

(iii) Elimination of water [1]

(iv)



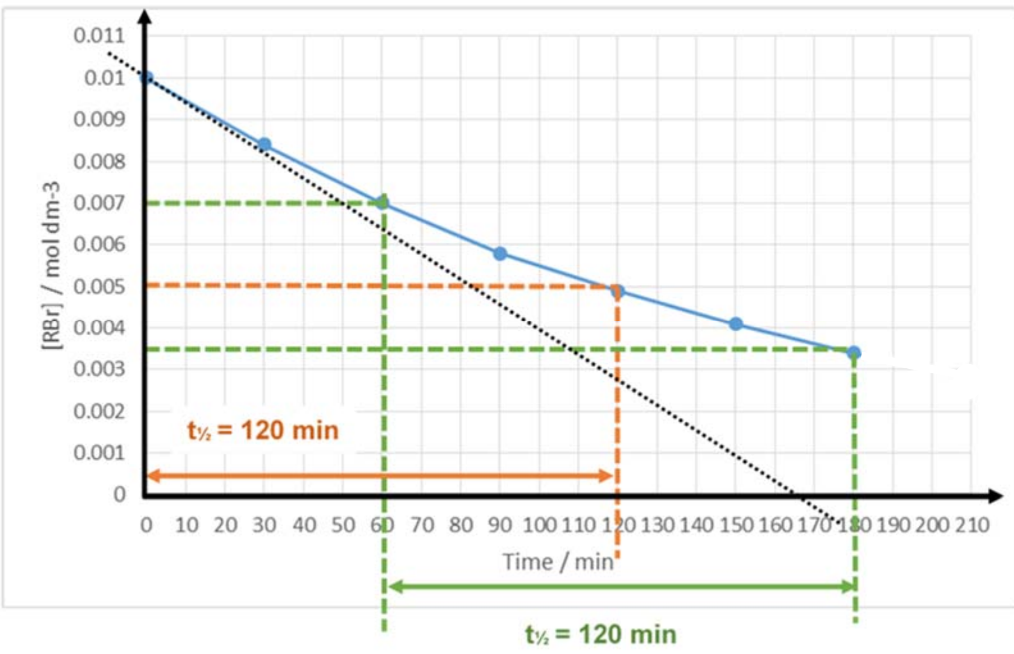
cis *trans*

Correct isomer and label [1] × 2 [2]

(v) Y is $CH_3CH{=}CHCO_2H$ [1]

(b) (i) So that [NaOH] remains relatively constant and any change in rate of reaction is only due to change in [bromoalkane] [1]

(ii)



Correct axes with labels and units + smooth curve [1]
Construction lines to obtain of at least two $t_{½}$ + read $t_{½}$ correctly [1]
Explain that two or more $t_{1/2}$ are constant hence first order w.r.t. bromoethane [1]

(iii) **Rate = *k* [RBr] [NaOH]** [1]

(iv) From graph, initial rate = $\left(\dfrac{0.0100}{165}\right)$ [1]

$= \mathbf{6.06 \times 10^{-5}}$ (mol dm^{-3} min^{-1}) (ignore units)

- Draw gradient at t=0 min to determine initial rate and obtain value of 10^{-5}

(v) $k = \dfrac{6.06\times10^{-5}}{0.01\times0.10} \quad \dfrac{\text{mol dm}^{-3}\text{ min}^{-1}}{\left(\text{mol dm}^{-3}\right)^2}$ [1]

$= \mathbf{0.0606\ mol^{-1}\ dm^{3}\ min^{-1}}$

- Correct k and units (allow ecf)

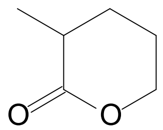
(c) **(i)** C–O bond of alcohol: <u>970–1260</u> (cm^{-1}) [1]
O–H bond of alcohol: <u>3580–3650</u> (cm^{-1})
C–H bond of alky group: 2850 – 2950 (cm^{-1})

[1]: any 2

(ii) C–O bond of carboxylic acids: 1210–1440 (cm^{-1}) [1]
C=O bond of carboxylic acids: 1680–1730 (cm^{-1})
O–H bond of carboxylic acids: 2500–3000 (cm^{-1})

[1]: any 1

(iii) **M** is [1]



Type of reaction: condensation / esterification

[1]